Neil Phan

hello@neil.place https://neil.place
Charlottesville, VA https://github.com/neil-phan

Education

University of Virginia, Charlottesville, VA May 2024
B.S. Computer Science, B.A. Mathematics GPA: 3.73/4.00
Relevant Courses: *Machine Learning, Computer Architecture, Algorithms, Probability, Linear Algebra, Computational Biology, Algorithmic Economics, Embedded Computing and Robotics, Data Structures*

Experience

Data Science Intern June 2022 – August 2022
Geospark Analytics

- Projects involving Time Series Analysis, Backend Development, Machine Learning Pipelines, Cloud Computing (AWS/Docker), and Natural Language Processing Models

Machine Learning Researcher January 2022 – May 2022
University of Virginia Security Research Group

- Increased the rate of convergence in model-targeted poison attacks by 43% in machine learning models by introducing modifying and removal of data points
- Analyzed papers on recent adversarial machine learning and machine unlearning discoveries

Data Structures and Algorithms Teaching Assistant January 2022 – May 2022
University of Virginia

- Held weekly office hours to help students in topics relating to data structures and algorithms
- Led weekly meetings with multiple cohorts to assist with lab assignments

Software Development Methods Teaching Assistant August 2021 – December 2021
University of Virginia

- Hosted weekly office hours to assist students on object-oriented programming
- Conducted bi-weekly labs to provide active learning and assist students in lab assignments

Math Instructor December 2019 – August 2021
Mathnasium

- Assisted 300+ K-12 students on various math topics ranging from addition to calculus
- Fostered 1-on-1 interactions with students to promote engagement and active learning

Projects

Evolution Simulator Using Neural Networks *Python, Numpy, Matplotlib, PyGame*

- Constructed a neural network that improves organisms' ability to find food each generation
- Designed randomized terrain generation to produce different environments for organisms

Liar's Dice AI with Q-Learning Agents *Python, Numpy, Matplotlib*

- Hashed and stored all 7776 possible Q-states the agent can hold with 61 unique bandit arms
- Ran 1 million training epochs to have a 70% agent win rate against human players

Fashion Image Classifier *Python, Keras, Tensorflow, Scikit-Learn, Numpy*

- Implemented a CNN to take in images and return the type of clothing with 92.6% accuracy
- Simplified neural network architecture to reduce training iteration time by 40%

Skills

Languages: Python, C++, Go, Java, SQL, HTML, CSS
Tools: PyTorch, Tensorflow, Scikit-Learn, Numpy, Pandas, Matplotlib, AWS, Docker, PostgreSQL